Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/07/2023
PI Name:	McLean, Robert Ph.D.		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Marcos	**State:**	TX
Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Per NASA-MSFC, the PI has received supplemental funding for an augmentation. End date changed to 08/31/2025 and grant number is now 80NSSC22K1361 (Ed., 4/1/25). NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 9/17/23). NOTE: End date changed to 08/31/2022 per NSSC information (Ed., 9/6/22). NOTE: End date changed to 12/4/2021 per NSSC information (Ed., 3/10/21)		
Key Personnel Changes/Previous PI:	September 2023 report: Key personnel on the original grant application are unchanged. This includes co-Principal Investigator (PI) Cheryl Nickerson and her team Jennifer Barrila (co-I) and Jiseon Yang (originally a postdoc, now Research Assistant Professor) at Arizona State University (ASU), as well as C Mark Ott (co-I) at NASA Johnson Space Center (NASA-JSC). Two additional participants, Sandhya Gangaraju and Richard Davis at Arizona State University (ASU) are also participating. Due to the expertise of our team, we performed all the necessary microscopy and will not require external collaborators as originally considered. In the PI's lab at Texas State (TxST), Starla Thornhill was a master's degree student as the grant was being written. She successfully defended her MS thesis in 2016 and completed her PhD in 2020 on work directly related to this grant. She continued as a postdoc during 2021 and along with Sandhya Gangaraju and Jiseon Yang (from ASU), completed all the postflight microscopy. As well, we have added Alistair McLean as an unpaid collaborator. He works as a software engineer and wrote a program that helps us enumerate		

	bacterial populations from microscopy images. Finally, as a result of discussions during the flight readiness review, we have been asked to consult with metallurgical experts from NASA Marshall Space Flight Center (NASA-MSFC) to assist our team in interpreting microbial corrosion information. We agree to this request as it will greatly enhance the science and benefits to NASA and our team. The specific metallurgical expert(s) are being identified by S. Gorti at NASA-MSFC.
COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, C. Mark Ph.D. (NASA Johnson Space Center) Nickerson, Cheryl Ph.D. (Arizona State University)
Grant/Contract No.:	80NSSC22K1361 ; NNX17AC79G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. As a result of supplemental funding, we are now able to gather genetic data on the mechanisms by which mixed bacterial species are able to form biofilms in microgravity conditions. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	September 2023 Update: Our spaceflight experiment launched on SpaceX CRS-21 (SpX-21) on December 6, 2020 and two of our three samples returned on January 14, 2021. The third sample returned with Crew-1 on May 2, 2021. We encountered two major issues that have led us to request a reflight (which has been approved). One issue was media contamination, rendering our data interpretation as inconclusive. A second issue (needlestick injury to a space crewmember) resulted in some of our samples incubating for 117 days on the International Space Station (ISS) rather than the planned 4 days. While that caused a potential loss of some transcriptome data (described in the Task Progress section), a really positive development from the extended incubation resulted in enhanced microbial corrosion data, as well as the return of viable microorganisms that can now be investigated for genetic changes as a consequence of prolonged spaceflight. We are now pursuing an active study of experimental evolution in collaboration with the microbiology lab at NASA Johnson Space Center (JSC) (Co-I Mark Ott, Sarah Wallace, and their team). Another major positive development is the initiative to develop a machine learning computational approach to identify microbial-induced corrosion (MIC). MIC represents an estimated annual global economic loss of $500B and detection is a key need. We anticipate that our analytical approach in this area will have major economic benefits, far beyond the space program. Once our reflight occurs, we will be able to complete all of our original objectives easily within a year of getting our samples.
Task Progress:	Our overall objectives for this proposal are to study surface-adherent bacterial growth (biofilm formation) on stainless steel materials used for the water recovery system on the International Space Station, with a mixed bacterial population and its potential contributions to stainless steel corrosion during spaceflight, and compare these effects to those seen on Earth-based studies (ground controls). Included in these studies were tests to investigate the role of a potential silver-based disinfectant in controlling biofilm formation and associated corrosion. At the outset, we proposed to study biofilms using light and electron microscopy during spaceflight as well as with identical experiments on Earth (ground controls). We have since expanded these techniques, and more importantly, the rigor of the microscopy investigations by developing a computer analysis approach for quantitative analysis of the light microscopy results. During the current year (2022-2023), we began the process of developing an artificial intelligence-based approach to measure microbial corrosion using electron microscopy. We also obtained approval from NASA to include analysis of genetic activity (officially termed transcriptome analysis) so that we would be able to determine the activities of the microorganisms during spaceflight and ground experiments in the presence and absence of silver disinfectants. Our initial flight experiment launched on SpaceX CRS-21 in December 2020, and most of our samples were returned in mid-January 2021. A later set of samples, containing living bacteria, returned in May 2021. Just prior to the December 2020 launch, we learned that one of the commercially-supplied ingredients of our bacterial growth media was contaminated, so while we obtained results from the initial experiments, we were unable to determine whether any effects were due to the spaceflight conditions or due to the contamination. As a result, we have been approved for a reflight, which is currently scheduled for March 2024. The goal of the reflight will be to conduct and analyze our original experiments related to biofilm growth, potential control by silver disinfection, and its role in microbial corrosion of stainless steel. One unexpected positive opportunity arose from our delayed sample (returned in May 2021) as we were able to culture living bacteria, that had been incubated in orbit for 117 days. We also have ground control bacteria that incubated for the same duration, and with the assistance of the microbiology team at NASA Johnson Space Center (JSC), we are able to sequence these organisms, as well as the original “ancestral bacteria” (cultures preserved prior to experimentation) and see whether spaceflight induces any unique mutations in these organisms, a scientific approach referred to as experimental evolution. These two studies (experimental evolution and developing artificial intelligence to diagnose microbial corrosion) have been our most active areas of research during the 2022-23 period. In summary, while the original contamination event and reflight scheduling has been quite challenging, we have obtained some unexpected bonus science in computer-aided image analysis of biofilms and corrosion, as well as an unexpected and highly meaningful opportunity to investigate whether spaceflight affects bacterial evolution. With a future successful reflight, we anticipate a highly successful scientific outcome.

Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Castro CL, Velez-Justiniano YA, Stahl-Rommel S, Nguyen HN, Almengor A, Dunbar B, McLean RJC, Sysoeva TA, Castro-Wallace SL. "Genome sequences of bacteria isolated from the International Space Station water systems." Microbiol Resour Announc. 2023 Jun 7:e0015823. https://doi.org/10.1128/mra.00158-23 ; PMID: 37284768; PMCID: PMC10353376 , Jun-2023
Articles in Peer-reviewed Journals	Goeres DM, Velez-Justiniano YA, Kjellerup BV, McLean RJC. "Biofilm and human spaceflight." Biofilm. 2023 Aug 18;6:100150. https://doi.org/10.1016/j.bioflm.2023.100150 ; PubMed PMID: 38196503; PubMed Central PMCID: PMC10774301 , Dec-2023
Articles in Peer-reviewed Journals	Ranson TM, Barton ME, McLean RJC. "Influence of central metabolism disruption on Escherichia coli biofilm formation." Can J Microbiol. 2023 Oct 18. https://doi.org/10.1139/cjm-2023-009 ; PMID: 37728257 , Oct-2023
Articles in Peer-reviewed Journals	Nickerson CA, McLean RJC, Barrila J, Yang J, Thornhill SG, Banken LL, Porterfield DM, Poste G, Pellis NR, Ott CM. "Microbiology of human spaceflight: Microbial responses to mechanical forces that impact health and habitat sustainability." Microbiol Mol Biol Rev. 2024 Sep 26;88(3):e0014423. Review. https://doi.org/10.1128/mmbr.00144-23 ; PMID: 39158275; PMCID: PMC11426028 , Sep-2024